

COSC 594/650

Lagrangian Duality

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From unconstrained to constrained optimization

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n \end{array}$$

$\Downarrow$

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in X \end{array}$$

QUESTION: How should we modify our existing algorithms?

Constrained linear regression

$$\begin{array}{ll}\underset{\theta}{\text{minimize}} & \|X\theta - y\|^2 \\ \text{subject to} & \theta \geq 0 \\ & \mathbf{1}^\top \theta = 1\end{array}$$

- ★ $[\theta]_+$ – projection onto a polyhedron, itself a quadratic program

Convex, but hard to project!

QUESTION: Can we move the constraints into the objective?

Primal problem and Lagrangian

Functions: $f_0, f_1, \dots, f_m: \mathbb{R}^n \rightarrow \mathbb{R}$ (Convexity is not required.)

Simple set: $X \subset \mathbb{R}^n$

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & x \in X \end{array}$$

■ LAGRANGIAN

$$L(x, \lambda) = f_0(x) + \lambda_1 f_1(x) + \dots + \lambda_m f_m(x)$$

- ★ $\lambda := (\lambda_1, \dots, \lambda_m) \geq 0$ – Lagrange multipliers, or dual variables
- ★ $\lambda_i f_i(x)$ – price for violating the i -th constraint

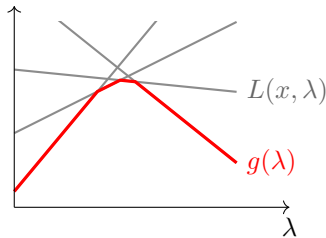
The dual function is always concave

Lagrangian: $L(x, \lambda) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x)$

■ DUAL FUNCTION

$$g(\lambda) = \inf_{x \in X} L(x, \lambda)$$

- ★ $L(x, \lambda)$ is affine in λ for each fixed x
- ★ $g(\lambda)$ is concave – pointwise infimum of affine functions



g is concave even if the primal problem is not convex

Lower bound property

Feasible point: $f_i(x) \leq 0$ for $i = 1, \dots, m$, and $x \in X$

Nonnegative multipliers: $\lambda \geq 0$

■ LOWER BOUND

$$\begin{aligned} f_0(x) &\geq f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) \\ &\geq \inf_{z \in X} \left(f_0(z) + \sum_{i=1}^m \lambda_i f_i(z) \right) \\ &= g(\lambda) \\ \implies g(\lambda) &\leq p^*, \quad \forall \lambda \geq 0 \end{aligned}$$

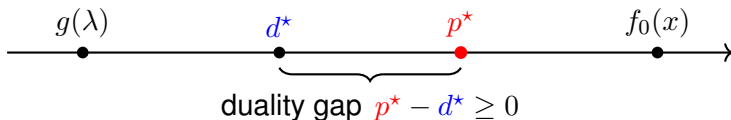
The Lagrange dual problem

■ DUAL PROBLEM

$$\begin{aligned} d^* &:= \underset{\lambda}{\text{maximize}} && g(\lambda) \\ &\text{subject to} && \lambda \geq 0 \end{aligned}$$

- ★ g is concave over $\lambda \geq 0$ – always a convex problem
- ★ d^* – best lower bound on p^*

■ WEAK DUALITY



Example: linear programming

$$\begin{array}{ll}\underset{x}{\text{minimize}} & c^\top x \\ \text{subject to} & Ax \leq b\end{array}$$

■ LAGRANGIAN

$$L(x, \lambda) = c^\top x + \lambda^\top (Ax - b) = \left(c + A^\top \lambda\right)^\top x - b^\top \lambda$$

■ DUAL FUNCTION

$$g(\lambda) := \inf_x L(x, \lambda) = \begin{cases} -b^\top \lambda, & \text{if } A^\top \lambda + c = 0 \\ -\infty, & \text{otherwise} \end{cases}$$

Dual problem:	maximize	$-b^\top \lambda$
	$\lambda \geq 0$	
	subject to	$A^\top \lambda + c = 0$

Example: quadratic programming

Positive definite matrix: $Q \succ 0$

$$\begin{array}{ll} \underset{x}{\text{minimize}} & x^\top Q x \\ \text{subject to} & Ax \leq b \end{array}$$

■ DUAL FUNCTION

$$\begin{aligned} \nabla_x L(x, \lambda) &= 2Qx + A^\top \lambda = 0 \\ \implies x^*(\lambda) &= -\frac{1}{2} Q^{-1} A^\top \lambda \\ \implies g(\lambda) &:= L(x^*(\lambda), \lambda) = -\frac{1}{4} \lambda^\top A Q^{-1} A^\top \lambda - b^\top \lambda \end{aligned}$$

Dual problem:

$$\underset{\lambda \geq 0}{\text{maximize}} \quad -\frac{1}{4} \lambda^\top A Q^{-1} A^\top \lambda - b^\top \lambda$$

Strong duality and Slater's condition

Convex problem: $f_0, f_1, \dots, f_m$ are convex, and X is convex

■ STRONG DUALITY

$$d^* = p^*$$

★ convexity + **constraint qualifications** – sufficient condition

■ SLATER'S CONDITION

$$\exists x \in \text{relint } X \quad \text{with} \quad f_i(x) < 0, \quad i = 1, \dots, m$$

★ strictly feasible point – depends only on the constraints

★ affine f_i , e.g., $1 \leq X\theta \leq 5$ – $f_i(x) \leq 0$ suffices

Convexity + Slater condition $\implies$ strong duality

Duality in algorithms

Primal feasible point: $x_k, \quad k = 0, 1, 2, \dots$

Dual feasible point: $\lambda_k \geq 0, \quad k = 0, 1, 2, \dots$

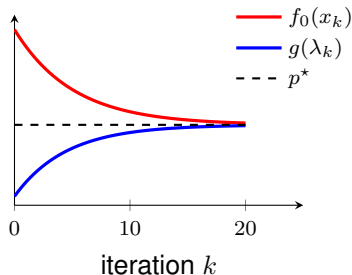
■ PRIMAL-DUAL GAP

$$g(\lambda_k) \leq p^* \leq f_0(x_k)$$

★ $f_0(x_k) - g(\lambda_k) \rightarrow 0$ – strong duality

■ STOPPING CRITERION

$$f_0(x_k) - g(\lambda_k) \leq \epsilon \implies f_0(x_k) - p^* \leq \epsilon$$



Accuracy is certified without knowing p^*

Questions?

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